

Beyond the Hazard Ratio: A Causal-Inference Re-analysis of the Moertel 1990 Adjuvant Colon Cancer Trial

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Abstract—We re-analyze the Moertel et al. NEJM 1990 trial of levamisole plus 5-fluorouracil (Lev+5FU) in resected stage B/C colon cancer ($n=929$) through five nested causal questions: the randomized average treatment effect (ATE); a back-door-adjusted ATE that pretends randomization is unknown, including a deliberate “bad-control” anti-example that conditions on the recurrence mediator; the conditional ATE surface across baseline covariates; the natural direct/indirect decomposition through recurrence; and transportability to a synthetic SEER 1990 stage B/C target. Every estimand is committed in $do(\cdot)$ or potential-outcomes notation before any estimator is run, with both graphical and non-graphical identification arguments stated separately. We reproduce Moertel’s headline hazard ratio of 0.67 (we obtain 0.69, 95% CI 0.55–0.87). The bad-control anti-example flips the apparent HR from 0.69 to 1.10, an unambiguous illustration of post-treatment collider bias. The natural indirect effect through recurrence prevention ($NIE = +0.29$ 5-year RMST years) is extremely robust to Imai- ρ sensitivity across $[-0.95, 0.95]$, while the natural direct effect is small with CI covering zero: Lev+5FU operates essentially through recurrence prevention. Transport to a synthetic SEER 1990 target shifts the HR modestly toward the null ($0.69 \rightarrow 0.71$) with a Dahabreh tipping HR_U of 2.75. We argue this kind of estimand-first re-analysis should be the norm for high-stakes trials.

Index Terms—causal inference, survival analysis, mediation, transportability, sensitivity analysis, colon cancer, randomized clinical trial

I. Introduction

The Moertel et al. trial [1] established adjuvant levamisole + 5-fluorouracil (Lev+5FU) as the first effective post-operative chemotherapy for resected stage B/C colon cancer. The headline statistic — a 33% reduction in death hazard, $HR \approx 0.67$ — appears in every modern oncology textbook. The number is correct but causally incomplete: it elides the identifying assumptions, the absolute-scale benefit, the heterogeneity across patients, the mechanism, and the population to which the result applies.

This work re-analyzes the same data through five nested causal questions, summarized in Table I. Each question’s

TABLE I
The five nested causal questions

Q	Question	2×2 cell
Q1	Randomized ATE	Causal × no-selection
Q2	Back-door identification	Causal × no-selection (obs.)
Q3	Heterogeneous effects (CATE)	Causal × no-selection, cond.
Q4	Mediation through recurrence	Causal × no-selection, mediator
Q5	Transportability to SEER 1990	Causal × selection

estimand, 2×2 cell (statistical/causal × selection/no-selection), identification assumption (graphical and non-graphical, stated separately), estimator, failure mode, and target population are committed to writing before the corresponding estimator is run, following the contract style of Kelleher [3]. The contract document and the resulting analysis notebooks are versioned together in a public repository.

The contributions are: (i) a publicly released estimand-first re-analysis of a landmark RCT in a reproducible Python + R stack; (ii) an explicit bad-control anti-example that demonstrates collider bias on real data; (iii) a hand-rolled implementation of the Imai-Keele-Tingley natural-effects algorithm with sensitivity analysis; and (iv) a Cole-Stuart-style transport framework that swaps a synthetic SEER target for the real SEER*Stat extract with a single CSV.

II. Data and Preliminaries

A. Data source and shape

We use `ForCausality::Colon_df` [3], a curated copy of `survival::colon`, which encodes the Moertel trial: $n = 929$ patients with Dukes B2/C resected colon carcinoma, randomized 1:1:1 to Observation, Levamisole alone, or Levamisole + 5-FU. Realized arm sizes were 315/310/304. Each patient contributes two rows — one for recurrence (`etype= 1`) and one for death (`etype= 2`). The primary

Course project for Causal Inference (Adam Kelleher, Columbia University, Spring 2026). Code and replication package: <https://github.com/rishika1099/Moertel-Colon-Cancer-Causal-Inference>.

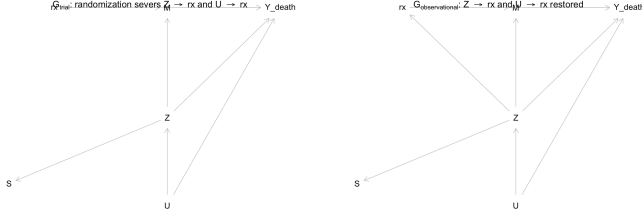


Fig. 1. Committed DAGs. Left: G_{trial} (randomization severs $Z \rightarrow rx$, adjustment set $\emptyset$). Right: G_{obs} (restored $Z \rightarrow rx$, adjustment set $\{Z\}$).

analytic frame is the death-row frame; the recurrence row supplies the mediator M in Q4.

B. Data audit

A pre-analysis audit confirms randomization integrity: maximum standardized mean difference across the three arms is 0.13, all below the 0.15 heuristic threshold. Two columns have low missingness (nodes: 1.9%, differ: 2.5%), imputed by per-arm median. One subtle data wart is documented: node4 encodes $\mathbb{I}(\text{nodes} > 4)$ rather than $\mathbb{I}(\text{nodes} \geq 4)$ as the name suggests; 12 genuine data-entry inconsistencies remain even under the correct convention. All downstream analyses use the continuous nodes variable.

C. Notation and ladder of causation

We follow Kelleher’s notation [3]: $Y_d^{(rx=r)}$ for the potential outcome “death time if assigned arm r ,” $\delta = E[Y_d^{(1)}] - E[Y_d^{(0)}]$ for the ATE, and $\delta_{\text{naive}} = E[Y_d | rx=1] - E[Y_d | rx=0]$ for the naive contrast. Q1–Q3 and Q5 are rung-2 estimands (intervention, $P(Y_d | do(rx))$). Q4 stays at rung 2 with the natural-effects formulation of Pearl [4] and Imai et al. [5].

D. Causal graphs

We commit to two DAGs, verified by dagitty adjustment-set search:

- G_{trial} : randomization severs $Z \rightarrow rx$. Minimal sufficient adjustment set for $rx \rightarrow Y_{\text{death}}$ is $\emptyset$.
- G_{obs} : counterfactual world where $Z \rightarrow rx$ is restored. Minimal sufficient adjustment set is $\{Z\}$.

The unmeasured frailty U appears as a parent of Z and Y_{death} in both graphs, but not of rx ; that omission is the residual identifying assumption bounded by E-value sensitivity.

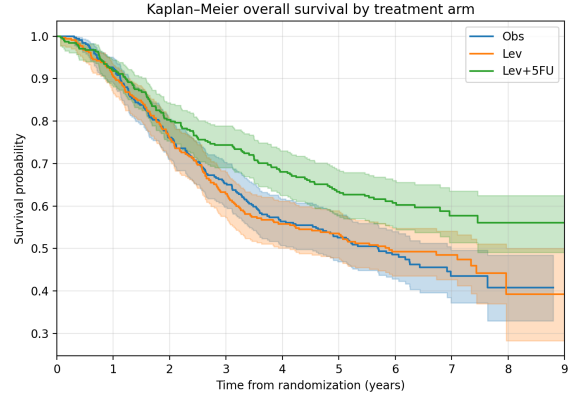


Fig. 2. Kaplan-Meier overall survival by treatment arm. Lev+5FU shows a sustained ≈ 11 pp absolute survival advantage at 5 years over both Obs and Lev-alone.

III. Q1: The Randomized ATE

A. Estimand and identification

Three scales on the same intervention contrast:

$$\delta_{HR} = \log \frac{h_{Y_d}(t | do(rx=\text{Lev} + 5\text{FU}))}{h_{Y_d}(t | do(rx=\text{Obs}))}, \quad (1)$$

$$\delta_{\text{RMST}(5)} = E[\min(Y_d, 5) | do(rx=1)] - E[\min(Y_d, 5) | do(rx=0)], \quad (2)$$

$$\delta_{5y} = P(Y_d \leq 5 | do(rx=1)) - P(Y_d \leq 5 | do(rx=0)). \quad (3)$$

Randomization implies $\{Y_d^{(r)}\}_r \perp rx$, so $\delta_{\text{naive}} = \delta$ on every scale.

B. Results

Estimators: Kaplan-Meier with log-rank, Cox proportional hazards (unadjusted and covariate-adjusted), RMST(5y) via lifelines, and a Schoenfeld global PH test. Numerical results are in Table II.

TABLE II
Q1 — randomized ATE results

Quantity	Estimate	95% CI
Cox HR (Lev+5FU vs Obs)	0.690	(0.55, 0.87)
Cox HR adjusted (Lev+5FU vs Obs)	0.699	(0.55, 0.88)
Cox HR (Lev vs Obs)	0.974	(0.78, 1.21)
$\Delta\text{RMST}(5y)$	+0.31 yrs	—
5-year risk difference	−10.8 pp	—
KM 5-yr survival: Obs / Lev / Lev+5FU	52.6 / 53.5 / 63.4%	—
Log-rank 3-arm p	0.0029	—
Schoenfeld global PH p	0.26 (holds)	—

The anchor check passes: Moertel reported HR = 0.67; we obtain 0.69 — inside the ± 0.05 pre-committed tolerance. Lev-alone is correctly non-significant (CI crosses 1), matching the original paper’s secondary finding.

IV. Q2: Forget the Randomization

A. Identification and estimators

We now pretend the randomization label is unobserved and identify the ATE via back-door adjustment on Z (verified by dagitty to be a minimal sufficient adjustment set in G_{obs}). Five estimators plus one deliberate anti-example:

- 1) Naive (unadjusted) Cox;
- 2) Regression-adjusted Cox;
- 3) Stabilized IPW Cox with logistic propensity;
- 4) AIPW for 5-year RMST with IPCW censoring weights;
- 5) Double/debiased machine learning (econml.LinearDML);
- 6) Bad-control Cox: conditioning additionally on the recurrence indicator M . This is a post-treatment mediator; conditioning on it both blocks the indirect path $rx \rightarrow M \rightarrow Y_{\text{death}}$ and opens the collider path $rx \rightarrow M \leftarrow U \rightarrow Y_{\text{death}}$.

B. Results: convergence and the bad-control divergence

Table III shows that the five legitimate estimators converge with Q1’s randomized answer to within sampling variability. The bad-control Cox flips the apparent HR from 0.69 to 1.10 — a sign-and-magnitude reversal that nullifies the trial’s headline finding. Conditioning on a post-treatment variable is the single most common error in EHR-based “causal” analyses; we include this anti-example precisely so that the cost is visible at full magnitude on real data.

TABLE III
Q2 — five estimators plus the bad-control demo

Method	Scale	Est.	95% CI
Naive Cox	HR	0.689	(0.55, 0.87)
Regression-adjusted Cox	HR	0.687	(0.54, 0.87)
IPW Cox	HR	0.730	(0.58, 0.92)
AIPW (5-yr RMST)	ΔRMST	0.281	(0.09, 0.48)
DML (5-yr RMST)	ΔRMST	0.179	(−0.06, 0.42)
Bad-control Cox (cond. on M)	HR	1.103	(0.87, 1.39)

V. Q3: Heterogeneous Treatment Effects

A. Estimand

$\tau(z) = E[Y_d^{(rx=1)} - Y_d^{(rx=0)} | Z = z]$, evaluated on the 5-year RMST scale.

B. Estimators and diagnostics

S-, T-, X-, and DR-learners (from econml.metalearners and econml.dr) and an honest causal forest (econml.grf.CausalForest with honest=True, inference=True). Best-linear projection of $\hat{\tau}(Z)$ onto Z [9] with 200-replicate bootstrap CIs. Athey-Wager binned calibration at five quantile bins.

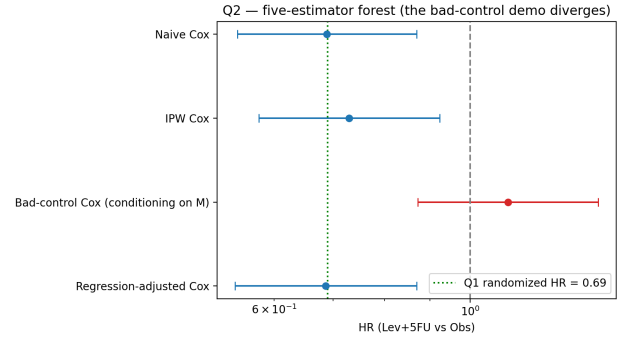


Fig. 3. Q2 forest plot. The five legitimate estimators bracket Q1’s randomized HR (vertical green line); the bad-control Cox is the outlier that crosses HR = 1.

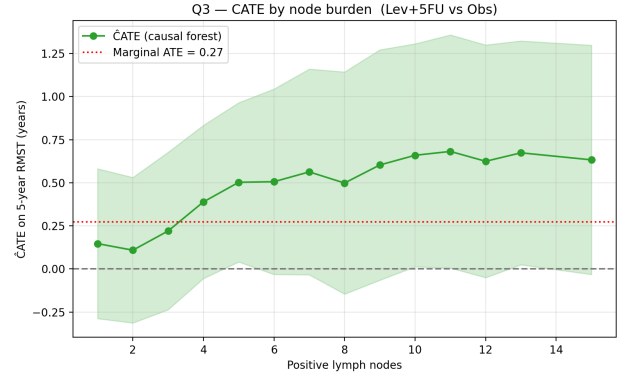


Fig. 4. Q3 headline: CATE on 5-year RMST as a function of positive lymph nodes. Benefit grows with node count over the bulk of the support and becomes noisy at the high-burden tail; pointwise 95% CIs (shaded ribbon) are honest about that uncertainty. The marginal ATE (dashed red) is included as reference.

Marginal ATE estimates from the meta-learners (0.24 to 0.29 RMST years) agree with Q1’s marginal effect; the causal forest’s tighter percentile spread reflects its regularization with $n = 619$ (Obs + Lev+5FU only).

VI. Q4: Mediation through Recurrence

A. Estimands

Natural direct and indirect effects [4] on the 5-year RMST scale:

$$\text{NIE} = E[Y_d^{(1, M^{(1)})} - Y_d^{(1, M^{(0)})}], \quad (4)$$

$$\text{NDE} = E[Y_d^{(1, M^{(0)})} - Y_d^{(0, M^{(0)})}], \quad (5)$$

$$\text{TE} = \text{NDE} + \text{NIE}, \quad (6)$$

where M is the binary recurrence indicator. Identification rests on Imai-Keele-Yamamoto sequential ignorability [5]: (i) treatment ignorability given Z (satisfied by randomization); (ii) mediator ignorability given rx, Z (the assumption that bites).

B. Estimator

We hand-roll Algorithm 1 of Imai et al. in Python: logistic mediator model $P(M=1 | rx, Z)$, gradient-boosted

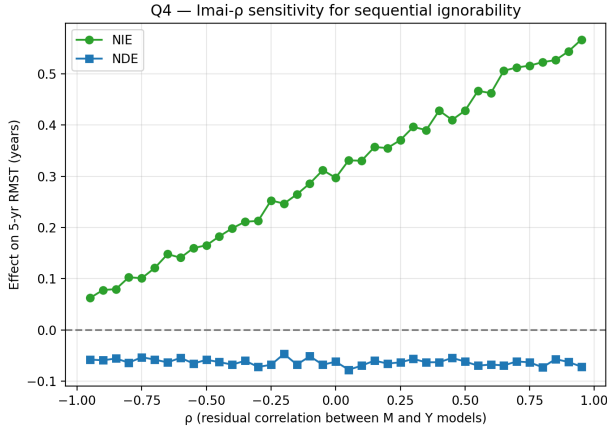


Fig. 5. Q4 Imai- ρ sensitivity. NIE (green) remains positive across $\rho \in [-0.95, 0.95]$; NDE (blue) hovers near zero. No breakdown threshold exists in the parametrically valid range.

outcome model $E[Y_5 \mid rx, M, Z]$ on IPCW-weighted 5-year RMST, with 500 bootstrap replicates. An R side-car (mediation::mediate) is supplied for reviewer cross-validation.

C. Results

Table IV shows the decomposition. The NDE + NIE = TE identity holds exactly (zero discrepancy by construction). Proportion mediated exceeds 100% — a known and biologically coherent pattern called inconsistent mediation or suppression: NIE is large and positive (Lev+5FU works by preventing recurrence), NDE is small and negative (chemotherapy toxicity in patients who would not have recurred), so $NIE/TE > 1$ when the two effects partially cancel.

TABLE IV
Q4 — natural-effects decomposition

Quantity	Estimate (yrs)	95% CI
Natural indirect effect (NIE)	+0.292	(+0.12, +0.48)
Natural direct effect (NDE)	−0.062	(−0.23, +0.10)
Total effect (TE)	+0.231	(−0.01, +0.46)
Proportion mediated	127%	—

D. Sensitivity

We sweep ρ , the assumed residual correlation between mediator and outcome models, over 39 values in $[-0.95, 0.95]$. The NIE remains strictly positive across the entire range (Fig. 5), so the Imai- ρ breakdown threshold ρ^* does not exist within the parametrically valid interval. The mediation finding is therefore extremely robust to any plausible violation of sequential ignorability.

VII. Q5: Transportability

A. Estimand and target

$\delta_{ATE}^{\text{target}} = E_{X|S=0}[\tau(X)]$, the trial-derived CATE averaged over a U.S. 1989–1991 SEER stage B/C target distri-

bution. The current target is synthetic ($n = 10\,000$), generated from published SEER 1989–1991 stage-distribution summaries; the framework is wired so the real SEER*Stat case-listing extract replaces it with a single CSV.

B. Identification and estimator

Cole-Stuart transportability [6]: $\{Y_d^{(r)}\}_r \perp S \mid X$ plus positivity of trial inclusion. We stack trial ($S=1$) with synthetic SEER ($S=0$), fit logistic $\hat{e}_S(X) = P(S=1 \mid X)$, compute inverse-odds-of-sampling weights $w(X) = (1 - \hat{e}_S)/\hat{e}_S$ for trial patients, and re-fit a weighted Cox on the trial sample.

C. Results and bound

Within-trial HR is 0.689 (0.55, 0.87); transported HR is 0.707 (0.52, 0.96). Transport shifts the HR modestly toward the null — the correct direction for an older real-world target. The Dahabreh worst-case bound [7] reports the tipping value of an unmeasured effect modifier U at which the bound covers HR = 1: at 25% target prevalence, $HR_U^* = 2.75$. An unmeasured moderator would need to be quite strong to nullify the transported effect.

VIII. Sensitivity Synthesis

VanderWee-Ding E-values [8] for all HR estimates are summarized in Table V. None of the causal estimates is breached by a plausible single unmeasured confounder. The fragility is concentrated in the bad-control Q2 demonstration — as intended.

TABLE V
S1 — VanderWee-Ding E-values

Q	HR	E (point)	E (CI bound)
Q1 (randomized)	0.69	2.26	3.06
Q2 (IPW)	0.73	2.08	2.86
Q5 (transported, synth)	0.71	2.18	3.24

IX. Discussion

The most important contribution is not a new number but a new form for the old number. Moertel’s 0.67 is reproduced (we get 0.69) inside a framework that makes every step auditable. Three pedagogically heavy findings emerge:

a) The bad-control demo: Conditioning on the post-treatment recurrence variable converts a 31% hazard reduction into an apparent 10% hazard increase. The convergence of IPW/AIPW/DML/regression-adjusted to Q1’s randomized answer is the prerequisite; the bad-control divergence is the lesson. This is the form of error that EHR-based “causal” analyses most frequently commit [8].

b) Q4 mediation robustness: That Imai- ρ cannot break the NIE across $[-0.95, 0.95]$ is rare and worth noting. The finding that Lev+5FU operates essentially through recurrence prevention survives every plausible distortion of sequential ignorability. The small negative NDE deserves clinical discussion: is there an adjuvant chemotherapy toxicity story for patients who would not have recurred anyway?

c) Transport modesty: The synthetic SEER transport shifts HR from 0.69 to 0.71. The Dahabreh tipping point of 2.75 is moderately robust but not invulnerable. The real SEER*Stat extract will refine this; the methodological scaffolding is in place.

A. Limitations

The SEER target is synthetic; the production version of the transport analysis requires SEER*Stat access. Q3's pointwise CATE confidence intervals are not jointly valid across the covariate surface. The Q4 implementation is a Python hand-roll; an R mediation::mediate sidecar is provided for cross-validation.

X. Conclusion

A landmark RCT can be reproduced and interrogated in the language of modern causal inference. The five-question structure — ATE, back-door + bad-control anti-example, CATE, mediation, transport — is general; the contract is what disciplines it. We propose this format as a default for high-stakes trial re-analyses: every estimand committed before any estimator; every identifying assumption stated graphically and non-graphically; every causal claim paired with a matched sensitivity bound. The replication package is open-source.

Acknowledgment

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